Cyclistic Bike-Share Analysis

How Do Annual Members and Casual Riders Use Cyclistic Bikes Differently?

Google Data Analytics Capstone — Case Study 1

Prepared by Kyle Pappas | 2019 Divvy Trip Data (Q1–Q4)

1. Business Task

Cyclistic's marketing team has expressed desire to convert casual riders into annual members, as members are more profitable and represent a stronger growth opportunity than continuing to broadly market to all riders. This analysis addresses the fundamental question of how do annual members and casual riders use Cyclistic bikes differently? This analysis attempts to define these behavioral differences will inform a future marketing strategy aimed at converting casual riders into members.

2. Data Sources

This analysis uses Cyclistic's historical trip data for all four quarters of 2019 (Q1–Q4), provided by Motivate International Inc. under license. Each quarterly file contains individual ride-level records including trip ID, start and end times, start and end station, bike ID, and rider type (member/casual). Combined, the four files represent approximately 3.8 million individual rides (Q1: 364,877 / Q2: 1,107,701 / Q3: 1,639,905 / Q4: 703,659).

The data is first-party and directly collected by Cyclistic's own bike-share system, making it a highly credible (ROCCC: Reliable, Original, Comprehensive, Current, Cited) source for this analysis. One limitation: due to data-privacy restrictions, the dataset does not include personally identifiable rider information, which means findings are limited to ride-level behavioral patterns rather than individual rider demographics or repeat-usage tracking across riders.

3. Data Cleaning & Preparation

Each quarterly dataset was reviewed and prepared in Excel by using Power Query prior to analysis. Two new fields were added to every record: ride length (the duration of each trip) and day of week (the day each trip began), enabling day-by-day and duration-based comparisons between rider types.

Each quarter was checked for common data quality issues, including missing values, duplicate trip records, and invalid station entries. No such issues were found in any quarter.

One data quality issue was identified and corrected: a small number of trips in each quarter (under 0.1% of records) showed ride durations exceeding 24 hours — in some cases, several weeks. These are consistent with bikes that were lost, stolen, or removed for maintenance rather than genuine customer trips, and were removed prior to analysis to avoid skewing average ride times. A total of approximately 1,650 such records were removed across the full year, out of roughly 3.8 million total trips.

Minor inconsistencies in file formatting between quarters (e.g., differing column names in the raw data) were also standardized to ensure the four quarters could be compared on equal terms.

4. Summary of Analysis & Key Findings

Analysis of the full year of 2019 trip data reveals consistent, distinct usage patterns between annual members and casual riders.

Member vs. Casual Riders Summary

Quarter	Casual %	Casual	Member	Total	Casual Avg	Member Avg
Q1 (Winter)	6.3%	23,095	341,782	364,877	35:17	11:19
Q2 (Spring)	23.4%	259,242	848,459	1,107,701	40:52	13:22
Q3 (Summer)	29.9%	491,045	1,148,860	1,639,905	39:42	13:49
Q4 (Fall)	15.1%	105,901	597,758	703,659	35:32	11:32
Annual Total	**23.0%**	**879,283**	**2,936,859**	**3,816,142**	**39:26**	**12:56**

Casual % = casual share of total rides. Casual / Member / Total = number of rides. Avg columns show mean ride length (mm:ss). The Annual Total row reflects full-year totals and volume-weighted averages, not a simple average of the four quarterly figures.

Ride Duration

Casual riders take significantly longer trips than members, averaging 39:26 (mm:ss) compared to members' 12:56 — roughly three times longer (see Appendix A for the quarterly breakdown). This gap remained stable across all four seasons, suggesting it reflects a fundamental difference in how each group uses the bikes, rather than a seasonal effect.

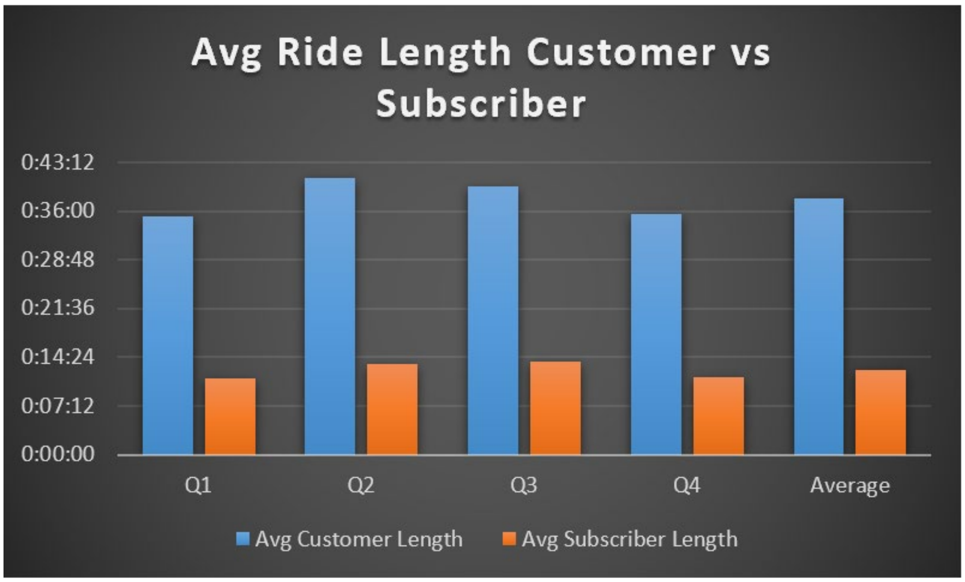

Chart 1. Average ride length by rider type and quarter (Customer = casual rider; Subscriber = annual member).

Day-of-Week Pattern

Members show a strong weekday usage pattern with a noticeable drop on weekends, consistent with commuting behavior. Casual riders show the opposite pattern, concentrating heavily on Saturdays and Sundays, consistent with recreational or leisure use (see Appendix D for the full annual breakdown).

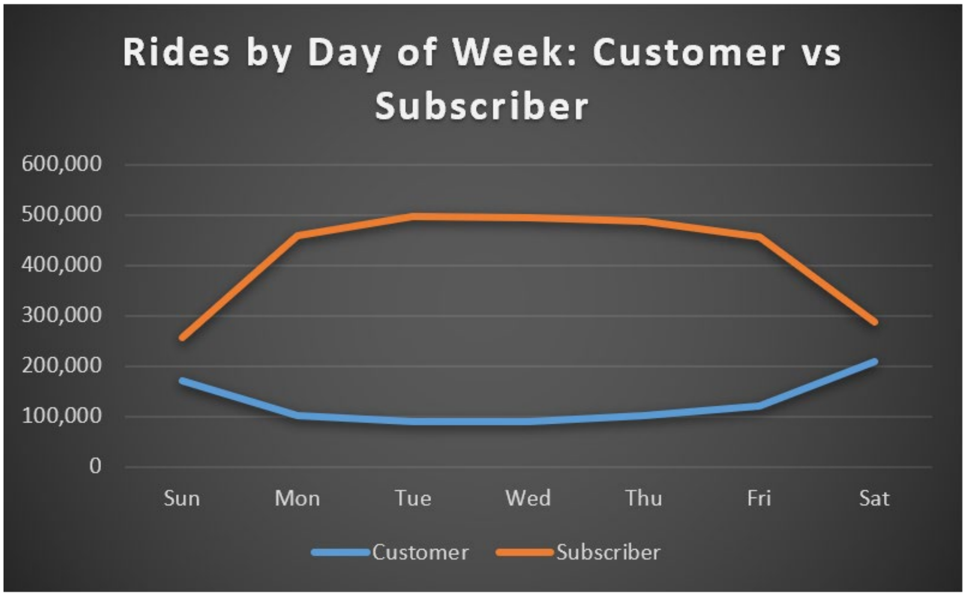

Chart 2. Ride counts by day of week and rider type (Customer = casual rider; Subscriber = annual member).

Seasonality

Casual ridership is highly seasonal, growing from just 6.3% of total rides in Q1 (winter) to a peak of 29.9% in Q3 (summer), before declining to 15.1% in Q4 (fall). Member ridership, while also increasing in warmer months, remains proportionally more stable year-round (see Appendix B).

Together, these patterns suggest that casual riders are primarily weekend leisure users whose activity is highly weather- and season-dependent, while members use Cyclistic bikes as a reliable, consistent transportation option throughout the year — most likely for commuting.

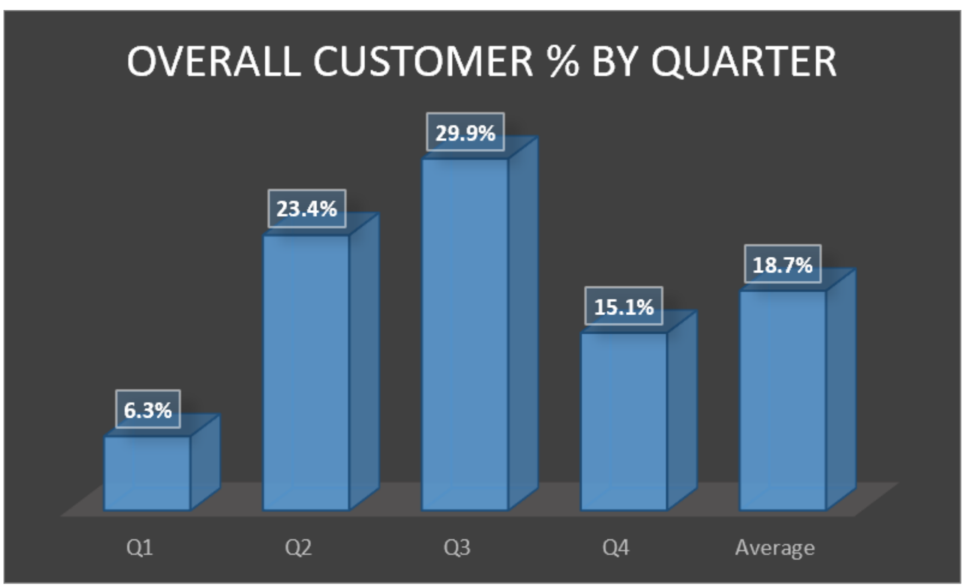

Chart 3. Casual rider (customer) share of total rides by quarter.

Overall Ride Volume by Rider Type

Across the full year, members account for the substantial majority of total ridership — 2,936,859 of 3,816,142 total rides (77.0%), compared to 879,283 casual rides (23.0%). Combined with the day-of-week pattern above, this indicates that Cyclistic's overall ride volume is driven predominantly by consistent, weekday commuter usage from members, rather than by casual, weekend-oriented use. While casual riders represent a meaningful and highly seasonal segment, members form the day-to-day backbone of Cyclistic's ridership.

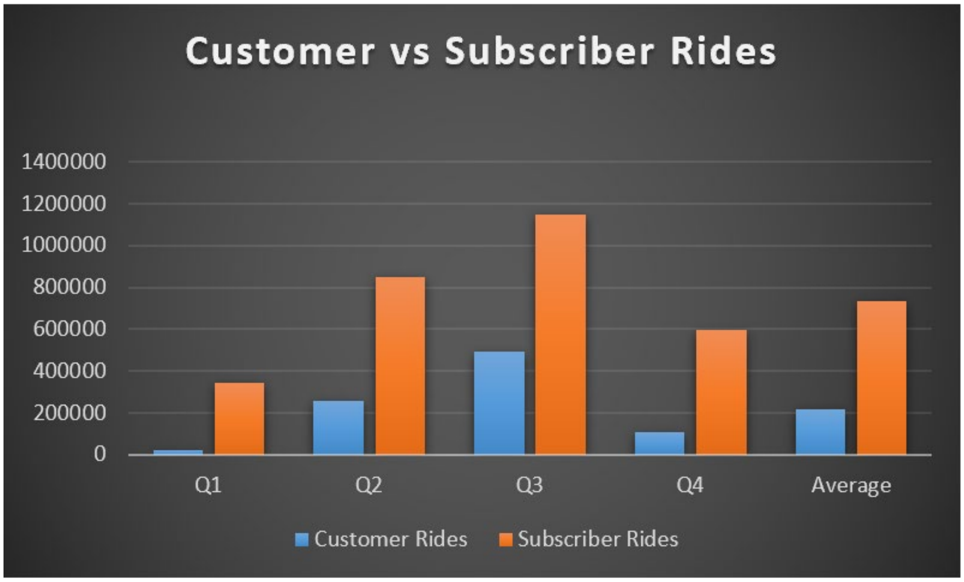

Chart 4. Ride volume by rider type and quarter (Customer = casual rider; Subscriber = annual member).

5. Top Three Recommendations

- Shift marketing messaging from “commute value” to “experience value” for casual riders. Data shows casual riders take trips averaging roughly three times longer than members (35–41 min vs. 11–14 min) and concentrate heavily on weekends — a leisure pattern, not a commuting one. Campaigns should be timed and worded around weekend recreational use (e.g., a prompt after a long weekend ride) rather than commuter-focused messaging aimed at members.
- Launch conversion campaigns in early spring (Q2), ahead of peak season. Casual ridership share nearly quadrupled between Q1 (6.3%) and Q2 (23.4%), continuing to climb into Q3’s peak (29.9%). Targeting riders as they are ramping up their activity — rather than waiting until they are already at peak summer engagement — gives Cyclistic a chance to capture a full summer-and-fall of membership value from a rider before their habits fully form around single-ride purchases.
- Introduce a mid-tier seasonal membership as a conversion bridge. Given the sharp seasonality of casual ridership, a full annual commitment may feel like a large leap for a rider who is only active a few months a year. A discounted seasonal pass (e.g., covering April–September) could lower the barrier to entry and serve as a stepping stone toward full annual conversion in a following year.

Appendix

Appendix A. Average Ride Length by Quarter (Detail)

Quarter	Casual Riders	Members
Q1 (Winter)	35:17	11:19
Q2 (Spring)	40:52	13:22
Q3 (Summer)	39:42	13:49
Q4 (Fall)	35:32	11:32
Annual (weighted avg.)	39:26	12:56

Annual figures are weighted by quarterly ride volume, not a simple average of the four quarterly figures.

Appendix B. Ride Volume & Casual Rider Share by Quarter (Detail)

Quarter	Casual Riders	Members	Total Rides	Casual Share
Q1 (Winter)	23,095	341,782	364,877	6.3%
Q2 (Spring)	259,242	848,459	1,107,701	23.4%
Q3 (Summer)	491,045	1,148,860	1,639,905	29.9%
Q4 (Fall)	105,901	597,758	703,659	15.1%
Annual Total	879,283	2,936,859	3,816,142	23.0%

Appendix C. Data Cleaning Log Summary

- Q1 2019: ~200 rows removed (ride length exceeding 24 hours).
- Q2 2019: 462 rows removed (ride length exceeding 24 hours); column names standardized to match Q1/Q3/Q4 convention.
- Q3 2019: ~700 rows removed (ride length exceeding 24 hours).
- Q4 2019: ~300 rows removed (ride length exceeding 24 hours).
- No blank values, duplicate trip IDs, sub-60-second rides, or test/maintenance station entries were found in any quarter.

Appendix D. Total Annual Rides by Day of Week (Detail)

Rider Type	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Casual	169,935	101,355	88,495	89,601	101,203	120,934	207,760
Member	256,184	458,714	496,951	494,205	486,837	456,890	287,078

Figures represent total ride counts summed across all four quarters of 2019.